Summary Report

1. Best-Performing Model and Why

- The best-performing model in this project is the **regularized linear model**, specifically **Ridge Regression**.
- Why:
    - It achieved the strongest validation performance among the tested candidates.
    - Ridge delivered lower validation error and higher R^2 than the unregularized linear baseline.
    - The model balanced predictive accuracy with generalization, making it the best choice for reliable pricing predictions.

2. Impact of Gradient Descent Optimization

- Gradient descent optimization was central to fitting the regression models and tuning the regularized linear estimators.
- Impact:
    - It allowed the model to converge efficiently toward the minimum loss when training on scaled features.
    - Optimization ensures stable coefficient updates and helps the model learn the underlying relationship between features and price.
    - When combined with regularization, it reduced the risk of divergence and produced more robust parameter estimates.

3. Evidence of Overfitting/Underfitting

- Evidence of overfitting:
    - The unregularized linear model showed a larger gap between training and validation error, indicating it fit training noise.
    - A less constrained model form would likely have exaggerated this gap further.
- Evidence of underfitting:
    - Simple linear models without enough features or regularization could underfit if they failed to capture the dataset's complexity.
    - However, the regularized Ridge model achieved high validation R^2, so underfitting was not the primary issue after tuning.

4. Practical business interpretation of results

- The model can be used by real estate professionals to estimate home values more consistently.
- Business implications:
    - **Pricing decisions**: supports setting more competitive and data-based listing prices.
    - **Investment guidance**: highlights which property features have the greatest influence on price.
    - **Risk mitigation**: the use of regularized regression makes predictions less sensitive to outliers and noisy data.
- Overall, the results support using a regularized regression approach as a dependable tool for market analysis and pricing strategy.